

COMP 10280 Programming I (Conversion)

Practical Sheet 7

Dé hAoine, 25 Meán Fómhair 2026

For each of the following questions, write an algorithm in pseudocode first before writing a Python program. Submit your algorithms in psuedocode as well as your Python programs.

1. Write a program that prompts the user for a year and checks whether the year is a leap year. Use my algorithm from Lecture 8. Ensure you understand how this program works and, in particular, how the conditions work.

Save this program as p7p1.py.

2. Write a program that prompts the user for a year and checks whether the year is a leap year. Use the algorithm on the Wikipedia page (also given in Lecture 8). Ensure you understand how this program works and, in particular, how the conditions work.

Save this program as p7p2.py.

3. Write a program that uses a `while` loop to go through the first 50 integers and prints out each number and the square of the number.

Save this program as p7p3.py.

4. Write a program that uses a `while` loop to sum the first 5000 integers and prints out the total.

Save this program as p7p4.py.

5. Write a program that uses a `while` loop to sum the integers in the range 1–10 000 that are divisible by 3 or by 5 and prints out the total.

Save this program as p7p5.py.

**Please upload your work to
the Brightspace site before Sunday
evening.**

**You should keep a copy of your programs
for your portfolio.**